

Framestore Farsight – A scalable virtual production ecosystem for faster iteration

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Figure 1: *The Last of Us*, 2023.

ABSTRACT

Real-time technology and game engines have fundamentally changed the film-making process, and will continue to disrupt the industry in the coming years. In this talk we will present *Farsight*, a virtual production ecosystem for faster and more collaborative visualisation than ever before. From concept to final pixel, we will demonstrate the virtual production workflows that have led to better communication, faster iteration and greater creativity.

CCS CONCEPTS

• **Computing methodologies** → **Motion capture; Mixed / augmented reality; Virtual reality.**

KEYWORDS

VFX, real-time, virtual production, Unreal Engine

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1 BACKGROUND

Pre-visualisation helps with the conceptualisation, planning and communication of complex VFX, and with the advancement of motion capture and real-time technology, this has become entirely interactive; enabling film-makers to interact with virtual worlds similarly to how they interact on set. Yet many of these new tools and workflows are disrupting a decades-old film-making process, presenting as many challenges as it does opportunities.

2 PRODUCTION REQUIREMENTS

Due to movement restrictions in the pandemic, *1899* relied heavily on the use of virtual production to develop virtual environments to be shot on a volume [Netflix Production Technology Resources 2022]. Working from concept art, our Virtual Art Department blocked out environments to be scouted in virtual reality. This led to the establishment of shots using virtual cameras and full-performance motion capture, which were then sent to shot creators and animators for the final rendered pre-vis. Once in production, this pre-vis is referenced on set, aiding the use of in-camera visual effects (ICVFX) and simulcam, which in turn both produce valuable reference for post-vis and post-VFX.

Our aim was to build an efficient end-to-end virtual production pipeline and tool-suite that would speed up iteration and collaboration throughout pre-production and production, capturing creative decisions that could be seamlessly integrated downstream.

3 IMPLEMENTATION

Farsight is a scalable virtual production ecosystem to scout, frame and visualise virtual content. With dedicated applications for augmented reality (AR) and virtual reality (VR), and close integration with the Unreal Editor, creative changes can be shared and iterated on across a variety of scenarios.

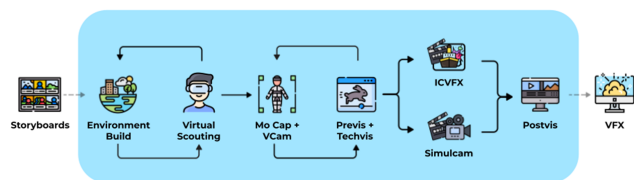


Figure 2: Virtual Production Workflow

3.1 Farsight VR

Farsight VR allows directors and production designers to scout virtual locations, experiment with different lenses, and leave feedback for artists directly in the environment; speeding up the communication and delivery of virtual set builds. These locations and cameras are then saved and restored downstream.

3.2 Farsight Studio

Farsight Studio is an Unreal Editor interface for operating motion capture, virtual cameras and ICVFX. In addition to speeding up the response time of our operators, we removed a significant bottleneck in sending virtual production data downstream by adding the ability to quickly review and export motion capture from within Unreal; automating the collection and organisation of virtual production data recorded across multiple machines. This data is automatically processed and made available in other 3D packages, with all associated reference ingested into ShotGrid.

3.3 Farsight Go

Farsight Go is our AR simulcam solution for iOS. With no external tracking or lens calibration needed, it provides a lightweight alternative to traditional simulcam solutions. By scanning surrounding geometry and sampling local lighting conditions, we are able to more accurately visualise what will be added in post-production, emulating different camera bodies and lenses to make better decisions on set, and provide valuable reference for post-vis and post-VFX.

3.4 Farsight Link

To provide end-to-end integration across all of our virtual production services, we needed to capture the creative changes in these different contexts and share them across the *Farsight* ecosystem.

Farsight Link is a data wrangling solution designed to capture and synchronise virtual production data across multiple sites, transposing the data where necessary for each platform. This unified pipeline is enabling us to bridge workflows more seamlessly; when a director places a camera in VR, the DOP can pick up from that camera location in a motion capture studio. If the position of digital characters change during a shoot, the exact layout can be restored in post-vis. When shooting on a volume, real camera movements are recorded and utilised in post-VFX.

4 CONCLUSIONS

By creating a seamless data flow, *Farsight* has successfully become a cornerstone of the pre-visualisation pipeline at Framestore, speeding up the iteration and development of environments and shots. Additionally, *Farsight* captures creative decisions made on set and



Figure 3: Farsight Go

provides invaluable 3D reference to artists during post-vis and post-VFX.

Faster and more accurate pre-visualisation is leading to more experimentation by reducing the cost of taking creative risks, yet conversely these same workflows are forcing film-makers to commit to creative decisions earlier in the process, especially in the case of ICVFX. As virtual production continues to play an ever-growing role in the film-making process, it is important to recognise how it is disrupting established workflows, and the impact this has on how different departments collaborate.

Virtual production workflows often happen in parallel, and managing the concurrent versions of assets across these contexts remains a challenge, however the asset management system designed for Framestore's Unreal Shot Engine offers promising results [Prado and Jones 2023], with more exploration needed to incorporate these workflows across virtual production as a whole.

By increasing the interoperability of our virtual production data, we can provide greater flexibility at any part of the film-making process, reducing technical barriers, and providing artists more time and freedom to create better art.

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